

C798

Informatics System Analysis and Design

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A hospital with a rooftop garden is replacing its nurse-scheduling system. The CIO wants to choose a vendor next week, but the team has not yet interviewed charge nurses or documented current scheduling rules. What should come next in the SDLC?

- A. Analysis: gather requirements from users and document current rules.
- B. Skip analysis, since every modern system meets all scheduling needs.
- C. Select the vendor now and gather requirements during setup.
- D. Begin staff training on a system.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

During requirements gathering for a telehealth platform at a rural health network with 7 clinics, stakeholders list their needs. Which is a non-functional requirement?

- A. Clinicians can schedule a video visit.
- B. Pages load within 2 seconds for 95% of requests.
- C. The system sends appointment reminders by text.
- D. The system should work well.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

Which requirement is functional?

- A. Stored data is encrypted.
- B. Every requirement is functional if users ask for it.
- C. A nurse can print discharge instructions in Spanish.
- D. The system is available 99.9% of the time.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A hospital sends lab results to a clinic's EHR. The message arrives and every field parses correctly, but the clinic's system cannot recognize the hospital's local code GLU-SER as a glucose test. Both organizations belong to the same state hospital association. Which level of interoperability is missing?

- A. Structural.
- B. Semantic.
- C. Foundational.
- D. Organizational.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Which standard is designed for exchanging health data through modern web-based (RESTful) APIs?

- A. Any PDF export, since all systems can read PDFs.
- B. LOINC.
- C. HL7 FHIR.
- D. ICD-10-CM.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

After go-live of new nursing flowsheets at a hospital with a new parking app, nurses complain that recording one set of vital signs takes 14 clicks. The vendor says the system meets every contract specification. What is the best next step?

- A. Run usability testing with nurses doing real tasks, measure time and clicks, and redesign.
- B. Have developers redesign the screens without nurse input.
- C. Provide more training only.
- D. Accept the design because it meets the contract.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A clinic with a busy walk-in schedule sets all workstations to log off automatically after 10 minutes of inactivity and encrypts its laptops. The clinic's lobby was recently remodeled. Under the HIPAA Security Rule, which type of safeguard are these?

- A. Privacy Rule requirements.
- B. Physical safeguards.
- C. Administrative safeguards.
- D. Technical safeguards.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

Which is an administrative safeguard under the HIPAA Security Rule?

- A. A locked server room.
- B. A periodic security risk analysis and workforce training.
- C. Encrypting all email, because encryption is always required.
- D. Unique user IDs for each staff member.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A new bed-management system costs \$600,000 up front and is expected to save \$150,000 a year in staff time. The hospital's cafeteria serves 2,000 meals a day. What is the simple payback period?

- A. 0.25 years
- B. It cannot be found without cafeteria costs.
- C. \$450,000
- D. 4 years

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A project must pay back its cost within 3 years and will save \$80,000 a year. What is the most it can cost up front?

- A. \$240,000
- B. \$26,667
- C. \$80,000
- D. \$160,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

Before go-live, a hospital with a newly certified stroke center has nurses run realistic admission scenarios in the new system to confirm it supports their work. Developers already tested each module on its own. What kind of testing are the nurses doing?

- A.** Regression testing.
- B.** Unit testing.
- C.** Integration testing.
- D.** User acceptance testing.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

An analyst at a 150-bed community hospital with a new MRI scanner is mapping how nurses currently hand off patients at shift change, including paper notes and verbal steps, before designing an electronic handoff tool. What is this activity?

- A.** Future-state design.
- B.** Cost-benefit analysis.
- C.** Current-state (as-is) workflow analysis.
- D.** Gap analysis.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A hospital ~~with a rooftop garden~~ is replacing its nurse-scheduling system. The CIO wants to choose a vendor next week, but the team has not yet interviewed charge nurses or documented current scheduling rules. What should come next in the SDLC?

Target word: next SDLC step **Decides:** Requirements have not been gathered.

A	Correct	Requirements must come before design or selection.
B	Absolute language	Local rules vary; no system fits all needs automatically.
C	Dropped element	Skips analysis; the choice may not fit real needs.
D	Inverted operation	Training belongs to implementation, near the end.

2. Answer B Concept 2

During requirements gathering for a telehealth platform ~~at a rural health network with 7 clinics~~, stakeholders list their needs. Which is a non-functional requirement?

Target word: non-functional requirement **Decides:** A measurable quality of performance.

A	Confusion set	A task the system performs: functional.
B	Correct	Performance is a quality attribute, a non-functional requirement.
C	Adjacent category	Another feature, so functional.
D	Dropped element	Drops the measurable criterion that makes a requirement testable.

3. Answer C Concept 2 · transfer

Which requirement is functional?

Target word: reverse: find the functional one **Decides:** A task the system performs.

A	Adjacent category	A security quality, usually non-functional.
B	Absolute language	Users request both kinds.
C	Correct	Describes a system function.
D	Confusion set	Availability is non-functional.

4. Answer B Concept 3

A hospital sends lab results to a clinic's EHR. The message arrives and every field parses correctly, but the clinic's system cannot recognize the hospital's local code GLU-SER as a glucose test. ~~Both organizations belong to the same state hospital association.~~ Which level of interoperability is missing?

Target word: missing level **Decides:** Format works but meaning is lost.

A	Confusion set	The fields parsed correctly, so structure works.
B	Correct	Shared codes, such as LOINC, preserve meaning.
C	Adjacent category	The message was delivered, so transport works.
D	Camouflage	Answers the association detail; the gap is in coding.

5. Answer C Concept 3 • transfer

Which standard is designed for exchanging health data through modern web-based (RESTful) APIs?

Target word: reverse: purpose given, name the standard **Decides:** API-based exchange.

A	Absolute language	PDFs are not computable data exchange.
B	Confusion set	A code system for lab tests and observations, not an exchange method.
C	Correct	FHIR defines resources exchanged through RESTful APIs.
D	Adjacent category	A diagnosis classification.

6. Answer A Concept 4

After go-live of new nursing flowsheets at a hospital ~~with a new parking app~~, nurses complain that recording one set of vital signs takes 14 clicks. The vendor says the system meets every contract specification. What is the best next step?

Target word: next step **Decides:** A usability complaint from real users.

A	Correct	User-centered testing finds and fixes the problem.
B	Inverted operation	Leaves out the users whose work is affected.
C	Adjacent category	Training cannot remove unnecessary clicks.
D	Dropped element	Drops user satisfaction; meeting specs is not the same as being usable.

7. Answer D Concept 5

A clinic ~~with a busy walk-in schedule~~ sets all workstations to log off automatically after 10 minutes of inactivity and encrypts its laptops. ~~The clinic's lobby was recently remodeled.~~ Under the HIPAA Security Rule, which type of safeguard are these?

Target word: safeguard type **Decides:** System-enforced controls.

A	Confusion set	The Privacy Rule governs uses and disclosures, not these controls.
B	Camouflage	Answers the lobby detail; physical covers facilities and devices.
C	Adjacent category	Administrative covers policies, training, and risk analysis.
D	Correct	Automatic logoff and encryption are technology controls.

8. Answer B Concept 5 • transfer

Which is an administrative safeguard under the HIPAA Security Rule?

Target word: reverse: find the administrative one **Decides:** A management or policy process.

A	Adjacent category	A physical safeguard.
B	Correct	Both are administrative safeguards.
C	Absolute language	Encryption is technical, and HIPAA treats it as addressable.
D	Confusion set	A technical safeguard.

9. Answer D Concept 6

A new bed-management system costs \$600,000 up front and is expected to save \$150,000 a year in staff time. ~~The hospital's cafeteria serves 2,000 meals a day.~~ What is the simple payback period?

Target word: payback period **Decides:** Cost ÷ annual savings.

A	Inverted operation	Divides savings by cost.
B	Camouflage	Cafeteria volume does not affect this project.
C	Confusion set	Cost left after one year of savings, not a time period.
D	Correct	$\$600,000 / \$150,000 = 4$ years.

10. Answer A Concept 6 • transfer

A project must pay back its cost within 3 years and will save \$80,000 a year. What is the most it can cost up front?

Target word: solve for cost **Decides:** Cost = savings × payback years.

A	Correct	$\$80,000 \times 3 = \$240,000$.
B	Inverted operation	Divides savings by years instead of multiplying.
C	Wrong stem number	One year of savings.
D	Dropped element	Counts only two of the three years.

11. Answer D Concept 7

Before go-live, a hospital ~~with a newly certified stroke center~~ has nurses run realistic admission scenarios in the new system to confirm it supports their work. ~~Developers already tested each module on its own.~~ What kind of testing are the nurses doing?

Target word: testing type **Decides:** End users confirming fit for their work.

A	Confusion set	Retests after changes to catch new breaks.
B	Camouflage	Describes the developers' earlier module tests.
C	Adjacent category	Checks that components work together, usually done by the technical team.
D	Correct	End users verify the system meets their needs.

12. Answer C Concept 8

An analyst at a 150-bed community hospital ~~with a new MRI scanner~~ is mapping how nurses currently hand off patients at shift change, including paper notes and verbal steps, before designing an electronic handoff tool. What is this activity?

Target word: activity name **Decides:** Documenting what happens now.

A	Inverted operation	Future state comes after the current state is understood.
B	Confusion set	Weighs costs against benefits, not workflow steps.
C	Correct	Maps the existing process, including informal steps.
D	Adjacent category	Compares current and future states once both are defined.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	5	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Analysis before design	The SDLC gathers user requirements and documents current processes before designing or buying a system.	Choosing a vendor before requirements exist.	Leaders want to pick a product before users are consulted.
2	Functional vs. non-functional requirements	Functional requirements state what the system does; non-functional requirements state how well it does it (speed, uptime, security, usability).	Calling any feature request non-functional, or any security need functional.	A measurable quality such as “within 2 seconds.”
3	Levels of interoperability	Foundational moves data; structural keeps its format; semantic keeps its meaning through shared codes; organizational aligns policy and trust. HL7 FHIR supports API-based exchange.	Assuming a message that parses correctly is fully interoperable.	Data arrives but the receiver cannot interpret a local code.
4	Usability and user satisfaction	Usability is judged by testing real users doing real tasks, measuring time, clicks, errors, and satisfaction.	Accepting a system because it meets contract specs.	Complaints about clicks or time after go-live.
5	HIPAA Security Rule safeguards	Administrative (risk analysis, training, policies), physical (facility and device protection), technical (access control, audit, encryption, automatic logoff).	Calling a system setting a physical or administrative safeguard.	Is it a policy, a locked space, or a system control?
6	Economic feasibility and payback	Simple payback period = up-front cost ÷ annual net savings.	Dividing savings by cost.	An up-front cost and a yearly savings figure.
7	Types of testing	Unit tests single parts; integration tests parts together; regression retests after changes; user acceptance testing has end users confirm the system supports their work.	Calling any clinician test an integration test.	Who is testing and against what.
8	Current-state workflow analysis	Map the as-is workflow, including informal steps, before designing the future state.	Designing the new tool from how the process is supposed to work.	Documenting what people do now, including paper and verbal steps.

Course scope (public description): In Informatics System Analysis and Design, a broad understanding of data systems is covered to build upon the Foundations in Nursing Informatics course. The importance of effective interoperability, functionality, data access, and user satisfaction are addressed. The student will be analyzing reports and integrating federal regulations, research principles, and principles of environmental health in the construction of a real-world systems analysis and design project. This course will be directly applicable to healthcare settings as electronic records management has become compulsory for healthcare providers. All of the information in this course will be directly tied to the delivery of quality patient care and patient safety. Foundations in Nursing Informatics is recommended as a prerequisite.